

EdgeBench: 134 real-world, day-long tasks across 6 capability families

Agent runtime ≥ 12 h per task · recorded human expert effort mean **57.2 h** · every task provides **rich, real-world feedback** for continuously learning from environment

Scientific & ML

Systems & SE

Optimization

Knowledge

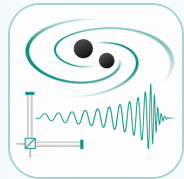
Formal

Games

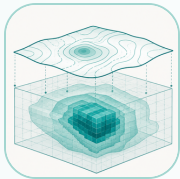
share of 134 tasks across 6 capability families

Scientific Problems & ML **39** tasks 29%

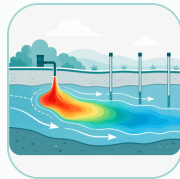
Inverse problems, forecasting & ML on real instrument/field data



Gravitational-wave detection



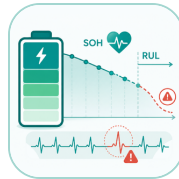
3-D gravity inversion



Groundwater plume modeling



Solar power forecasting



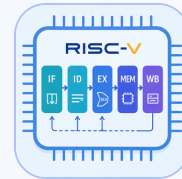
Battery health forecasting

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+34 more

Systems & Software Engineering **36** tasks 27%

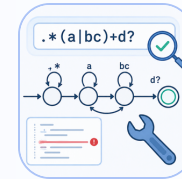
Large-scale changes to real production codebases



RISC-V CPU design



Matching-engine optimization



Regex engine repair



PocketBase development



TLS 1.3 implementation

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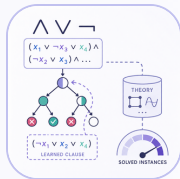
+31 more

Combinatorial Optimization **19** tasks 14%

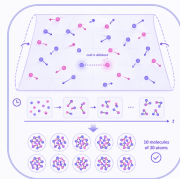
Open-ended optimization, judged on quality



Vehicle routing



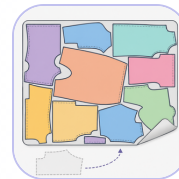
SAT / SMT solving



Molecular self-assembly



Job-shop scheduling



2-D irregular nesting

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+14 more

Professional Knowledge Work **19** tasks 14%

White-collar deliverables graded by client-style rubrics



CTA risk budgeting



Cross-border compliance



Claim-ring fraud audit



AIGC storyboarding



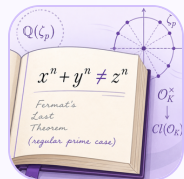
Brand annual planning

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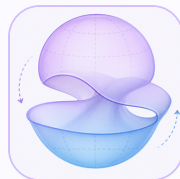
+14 more

Formal Math & Theorem Proving **13** tasks 10%

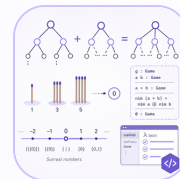
Machine-checked proving in Lean 4 & Coq



Fermat (regular case)



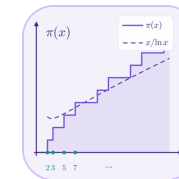
Sphere eversion



Combinatorial games



Erdős–Graham problem



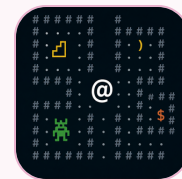
Prime Number Theorem

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+8 more

Interactive Games & Simulators **8** tasks 6%

Sequential decision-making in OOD worlds



NetHack



Dungeon Crawl



Transport tycoon sim



Text adventures



Wesnoth

...

+3 more